

LexLogica

SAT Mathematics

Formulas and knowledge, explained

47 core topics • Worked examples • Common mistakes

How to use this guide

Read the explanation, cover the example and solve it yourself, then check the common mistake. Use the formula together with its conditions. This reference supports practice across the core SAT Math domains; it is not a guarantee that every possible problem is represented.

Algebra — 8 topics

Advanced Math — 14 topics

Problem-Solving and Data Analysis — 12 topics

Geometry and Trigonometry — 10 topics

Problem-solving toolkit — 3 topics

Original LexLogica study material. Topic scope checked against College Board's published SAT Math domains on October 2, 2026. Independent material, not an official College Board publication.

Algebra

Linear equations and equivalent expressions

$$a(x + b) = ax + ab; ax + b = c \rightarrow x = (c - b)/a, a \neq 0$$

Distribute, combine like terms, and perform the same operation on both sides. In a word problem, define the variable and translate the relationship before calculating.

Example: $3(x - 2) + 4 = 16 \rightarrow 3x - 2 = 16 \rightarrow x = 6$.

Watch for: A minus before parentheses changes every sign. Do not divide by an expression that may be zero without considering that case.

Number of solutions and unknown constants

$$ax + b = cx + d \rightarrow (a - c)x = d - b$$

If $a \neq c$, there is one solution. If $a = c$ and $b = d$, every real x works. If $a = c$ and $b \neq d$, there is no solution. Use these conditions to find an unknown parameter.

Example: $2x + 3 = 2x + 5$ gives $3 = 5$: no solution. For $kx + 3 = 2x + 3$ to have infinitely many solutions, $k = 2$.

Watch for: An identity is true for every allowed input; a contradiction is true for none.

Rearranging formulas

$$A = bh/2 \rightarrow h = 2A/b, b \neq 0$$

Treat all other letters as constants while isolating the requested quantity. Keep grouped sums together.

Example: $C = 5(F - 32)/9 \rightarrow F = 9C/5 + 32$. At $C = 20$, $F = 68$.

Watch for: Dividing by a sum means dividing by the entire sum, not just one term.

Slope and linear functions

$$m = (y_2 - y_1)/(x_2 - x_1); y = mx + b; y - y_1 = m(x - x_1)$$

Slope is output change per input unit; b is the output when $x = 0$. Point-slope form constructs a line from one point and its slope. A constant first difference for equally spaced x -values indicates a linear model.

Example: Through $(2, 5)$ and $(6, 13)$, $m = 8/4 = 2$. Then $5 = 2(2) + b$ gives $b = 1$.

Watch for: A vertical line $x = c$ has undefined slope; a horizontal line $y = c$ has slope 0. A context may exclude $x = 0$ even when b exists algebraically.

Intercepts, parallel and perpendicular lines

x-intercept: set $y = 0$; y-intercept: set $x = 0$; perpendicular nonvertical slopes: $m_1 m_2 = -1$

Parallel distinct nonvertical lines have equal slopes and different intercepts. Perpendicular slopes are negative reciprocals; horizontal and vertical lines form the special case.

Example: For $2x + 3y = 12$, intercepts are $(6, 0)$ and $(0, 4)$. Its slope is $-2/3$, so a perpendicular line has slope $3/2$.

Watch for: An intercept is a point; the x- or y-coordinate alone is often what the question asks for.

Systems of two linear equations

Solve by substitution or elimination; graph intersections satisfy both equations

Multiply equations to cancel a variable, or isolate one variable and substitute. One intersection means one solution; parallel distinct lines mean none; identical lines mean infinitely many.

Example: $x + y = 7$ and $2x - y = 2$ add to $3x = 9$, so $(x, y) = (3, 4)$.

Watch for: When multiplying an equation, multiply every term, including the constant. Check both original equations.

Inequalities and solution regions

If $a < b$ and $c < 0$, then $ac > bc$

Reverse the inequality when multiplying or dividing by a negative number. In two variables, graph the boundary and test a point; a system uses the overlap of the regions.

Example: $-2x + 3 > 11 \rightarrow -2x > 8 \rightarrow x < -4$. For $y \leq 2x + 1$, include the boundary and shade below it.

Watch for: Strict inequalities use a dashed boundary; $\leq$ and $\geq$ include it. An “and” statement needs both conditions; “or” needs at least one.

Linear modeling and units

total cost = fixed cost + (cost per item)(number of items)

Define quantities with units. A coefficient is a rate; a constant is an initial or fixed amount. Tables, graphs, and equations can describe the same relationship.

Example: A rental costs \$15 plus \$4 per hour: $C = 15 + 4h$. A \$35 budget gives $h \leq 5$.

Watch for: Do not reverse input and output units. A maximum budget or capacity usually creates an inequality.

Advanced Math

Exponent laws and scientific notation

$$a^m a^n = a^{m+n}; a^m / a^n = a^{m-n}; (a^m)^n = a^{mn}; a^{-n} = 1/a^n; a^0 = 1$$

These rules simplify products, quotients, and powers. Negative and zero exponent rules require a nonzero base. Scientific notation is $c \times 10^n$, with $1 \leq |c| < 10$.

Example: $(3 \times 10^4)(2 \times 10^{-2}) = 6 \times 10^2 = 600$. $x^3/x^5 = 1/x^2$ for $x \neq 0$.

Watch for: Exponents do not distribute across addition: $(x + y)^2$ is not $x^2 + y^2$.

Radicals and rational exponents

$$a^{(m/n)} = (\sqrt[n]{a})^m; \sqrt{a^2} = |a|$$

For positive a , rational exponent rules work directly. Real even roots require a nonnegative radicand; odd roots allow negatives. Factor perfect squares to simplify radicals.

Example: $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$; $27^{(2/3)} = 3^2 = 9$.

Watch for: The square-root symbol means the nonnegative root. The equation $x^2 = 9$ has two solutions, but $\sqrt{9} = 3$.

Polynomial operations and factoring

$$a^2 - b^2 = (a - b)(a + b); (a \pm b)^2 = a^2 \pm 2ab + b^2$$

Combine only like powers; distribute all terms when multiplying. Factor out a greatest common factor first. In a monic quadratic $x^2 + bx + c$, find numbers with sum b and product c .

Example: $x^2 - 5x + 6 = (x - 2)(x - 3)$. Also $2x^2 + 7x + 3 = (2x + 1)(x + 3)$.

Watch for: Cancel common factors, not terms separated by plus or minus signs.

Quadratic equations and the quadratic formula

$$ax^2 + bx + c = 0 \rightarrow x = [-b \pm \sqrt{b^2 - 4ac}]/(2a), a \neq 0$$

Use factoring, completing the square, or the formula. The zero-product rule says a product is zero only when at least one factor is zero.

Example: $x^2 - 5x + 6 = 0 \rightarrow (x - 2)(x - 3) = 0 \rightarrow x = 2$ or 3 .

Watch for: Set the equation equal to zero before applying the formula. The denominator $2a$ applies to the whole numerator.

Discriminant and root relationships

$$D = b^2 - 4ac; r_1 + r_2 = -b/a; r_1 r_2 = c/a$$

$D > 0$ gives two distinct real roots, $D = 0$ one repeated real root, and $D < 0$ no real roots. Root sum and product follow by expanding $a(x - r_1)(x - r_2)$.

Example: For $x^2 + 6x + k = 0$ to have exactly one real solution, $36 - 4k = 0$, so $k = 9$.

Watch for: "One real solution" counts the repeated root once. Root relationships require standard form with nonzero a .

Quadratic graphs and completing the square

$$y = a(x - h)^2 + k; h = -b/(2a); k = f(h)$$

The vertex is (h, k) . If $a > 0$, k is the minimum; if $a < 0$, it is the maximum. In factored form $a(x - r_1)(x - r_2)$, the roots are visible.

Example: $x^2 - 6x + 11 = (x - 3)^2 + 2$, so the vertex is $(3, 2)$ and the minimum is 2.

Watch for: The vertex input is h , not $-h$. A contextual domain can make an endpoint the relevant maximum or minimum instead.

Functions, notation, domain and range

$$f(a) \text{ means the output at input } a; \text{ average rate} = [f(b) - f(a)]/(b - a)$$

Substitute into the entire rule. Domain is the set of allowed inputs; range is the set of resulting outputs. Read zeros, intercepts, extrema, and intervals of increase from a graph.

Example: If $f(x) = 2x^2 - 1$, then $f(3) = 17$. For $g(x) = \sqrt{x - 4}$, real inputs satisfy $x \geq 4$ and outputs satisfy $g(x) \geq 0$.

Watch for: $f(x)$ is not multiplication by f . A graph may be restricted by the real-world context.

Function transformations

$$g(x) = a f(x - h) + k$$

h shifts the graph right; k shifts it up. A negative a reflects across the x -axis; $|a|$ scales vertical distances. Replacing x by $-x$ reflects across the y -axis.

Example: $g(x) = -2(x - 3)^2 + 5$ has vertex $(3, 5)$, opens downward, and has a vertical scale factor 2.

Watch for: Inside changes act in the opposite direction: $f(x + 3)$ shifts left 3. $f(2x)$ compresses horizontal distances by $1/2$.

Exponential growth and decay

$$y = A(1 + r)^t; y = A(1 - r)^t; y = A b^{(t/k)}$$

A is the initial quantity; r is a decimal rate per period. In the last form, b is the multiplier every k time units. Equal input increments produce a constant output ratio.

Example: A population of 500 grows 8% yearly: $P = 500(1.08)^t$. After 2 years, $P = 583.2$. Doubling every 3 years uses $A \times 2^{(t/3)}$.

Watch for: Match the period to the time unit. A multiplier 1.08 is 8% growth, not 108% growth. Linear change adds a fixed amount; exponential change multiplies.

Solving exponential equations

$$\text{If } b^u = b^v \text{ with } b > 0 \text{ and } b \neq 1, \text{ then } u = v$$

Rewrite both sides using a common positive base when possible. For a context, a graph can identify the time when an exponential model reaches a target.

$$\text{Example: } 4^x = 8 \rightarrow 2^{(2x)} = 2^3 \rightarrow x = 3/2.$$

Watch for: Do not equate exponents when bases differ. Check whether the question asks for elapsed time, an exponent, or the quantity itself.

Rational expressions and equations

$$a/b + c/d = (ad + bc)/(bd), b, d \neq 0$$

Record denominator restrictions before simplifying. Factor, use a common denominator, and multiply an equation by its least common denominator to clear fractions.

$$\text{Example: } (x^2 - 9)/(x - 3) = x + 3 \text{ only for } x \neq 3. \text{ For } 2/(x - 1) = 3, x = 5/3.$$

Watch for: A canceled denominator still excludes its original zeros. Reject any candidate that makes an original denominator zero.

Radical and absolute-value equations

$$\sqrt{u} = v \text{ requires } u \geq 0 \text{ and } v \geq 0; |u| = c \text{ gives } u = \pm c \text{ if } c \geq 0$$

Isolate a radical before squaring, then check in the original equation. Absolute value is distance from zero; $|x - a|$ is distance from a.

$$\text{Example: } \sqrt{x + 2} = x \text{ gives } x^2 - x - 2 = 0. \text{ Candidates 2 and } -1 \text{ arise, but only 2 works. } |2x - 3| = 5 \text{ gives } x = 4 \text{ or } -1.$$

Watch for: Squaring can introduce false solutions. $|u|$ cannot equal a negative number.

Nonlinear systems

At an intersection, both equations hold

Use substitution to reduce a line-parabola or other nonlinear system to one equation. Solve for every allowed input, then recover the corresponding output.

Example: $y = x^2$ and $y = x + 2$ give $x^2 - x - 2 = 0$, so intersections are $(2, 4)$ and $(-1, 1)$.

Watch for: A tangency gives one intersection. Do not report only x-values if ordered pairs are requested.

Polynomial zeros and equivalent forms

$P(r) = 0 \Leftrightarrow (x - r)$ is a factor of polynomial $P(x)$

Factored form reveals zeros. Standard form reveals coefficients and the constant term. Rewriting an identity lets you compare matching coefficients.

Example: If $x^2 + kx + 12 = (x + 3)(x + 4)$ for every x , expanding the right gives $k = 7$.

Watch for: An equality for one particular x does not let you equate all coefficients. A repeated factor may touch the axis rather than cross it.

Problem-Solving and Data Analysis

Ratios, proportions and rates

$$a/b = c/d \rightarrow ad = bc; \text{ distance} = \text{rate} \times \text{time}; y = kx$$

A ratio compares quantities; a unit rate compares one unit of input. Direct proportion means a constant ratio y/x . Keep quantities in the same order.

Example: A mixture has red:blue = 2:3 and total 40. There are 5 ratio parts, each worth 8, so red = 16 and blue = 24.

Watch for: The fraction of red is $2/5$, not $2/3$. Average speed is total distance divided by total time, not usually the mean of speeds.

Unit conversions and density

$$\text{density} = \text{mass/volume}; \text{ population density} = \text{population/area}$$

Multiply by conversion factors arranged so unwanted units cancel. Square a length conversion for area and cube it for volume.

Example: $3 \text{ m}^2 = 3 \times 100^2 \text{ cm}^2 = 30,000 \text{ cm}^2$. A 120 g object occupying 40 cm^3 has density 3 g/cm^3 .

Watch for: 1 m^2 is not 100 cm^2 . Always include units in intermediate work.

Percentages and percent change

$$\text{part} = \text{whole} \times p/100; \text{ percent change} = 100(\text{new} - \text{old})/\text{old}$$

The reference amount is the denominator. Apply successive increases or decreases by multiplying their factors. Reverse a change by dividing by the factor.

Example: After a 20% discount, an item costs \$64: $\text{original} = 64/0.8 = \80 . A 20% rise then a 20% fall multiplies by $1.2 \times 0.8 = 0.96$.

Watch for: Percent change is different from percentage-point change: 30% to 40% is 10 points, but a $33\frac{1}{3}\%$ relative increase.

Mean and weighted mean

$$\text{mean} = \text{sum}/n; \text{ weighted mean} = \Sigma(\text{weight} \times \text{value})/\Sigma \text{weights}$$

Use frequencies as weights. A combined mean must account for different group sizes. The total equals the mean multiplied by the number of observations.

Example: Ten scores averaging 80 and twenty averaging 95 combine to $(10 \times 80 + 20 \times 95)/30 = 90$.

Watch for: A simple average of group means is valid only when their weights are equal.

Median, quartiles and spread

range = maximum – minimum; IQR = $Q_3 - Q_1$

Sort values before finding the median. For an even count, average the two middle values. Quartiles describe the middle half; standard deviation describes spread around the mean. Compare distributions and outlier sensitivity.

Example: For 2, 3, 4, 5, 16, the median is 4, mean is 6, and range is 14. The large value raises the mean more than the median.

Watch for: A box plot does not show every individual value or determine the exact mean. Use the given quartile convention when calculation is requested.

Transforming data

Add c : mean and median increase by c ; spread stays the same

Multiply by k : mean multiplies by k ; standard deviation multiplies by $|k|$

Adding a constant shifts all observations equally. Multiplying changes distances by the absolute scale factor. Range and IQR also scale by $|k|$.

Example: If every measurement increases by 10, its standard deviation is unchanged. If every measurement doubles, its standard deviation doubles.

Watch for: Changing only one observation is not the same as shifting the whole data set. More spread means larger standard deviation, not necessarily a larger mean.

Frequency tables, histograms and box plots

relative frequency = category count/total count

Read axes and whether entries represent counts, proportions, or percentages. Histograms group numerical values into intervals; a box plot shows a five-number summary.

Example: If 12 of 40 students fall in a category, its relative frequency is $0.30 = 30\%$.

Watch for: A histogram interval does not reveal each exact value. Do not infer exact individual scores or means from grouped values without sufficient information.

Scatterplots, models and residuals

linear prediction: $\hat{y} = mx + b$; residual = observed y – predicted $\hat{y}$

Interpret direction, strength, and form of association. Choose a linear, quadratic, or exponential model based on the pattern. A residual measures prediction error.

Example: A model predicts 70 while the observed value is 74, so the residual is +4: the point lies above the model.

Watch for: Association alone does not prove causation. Predictions far outside the observed input range are less reliable.

Probability and complements

$P(A) = \text{favorable/total (equally likely outcomes)}; P(\text{not } A) = 1 - P(A)$

Probabilities range from 0 to 1. Use counts or proportions from tables. For mutually exclusive events, probabilities add. In general $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.

Example: If 7 of 20 equally likely outcomes are favorable, $P(A) = 7/20$ and $P(\text{not } A) = 13/20$.

Watch for: Do not count overlapping outcomes twice. "Or" normally includes outcomes in both events.

Conditional probability and independence

$P(A \text{ given } B) = P(A \text{ and } B)/P(B), P(B) > 0$

Restrict the sample space to B before calculating. Independent events satisfy $P(A \text{ and } B) = P(A)P(B)$. Sampling without replacement usually changes later probabilities.

Example: Among 30 students who take art, 12 also take music. $P(\text{music given art}) = 12/30 = 0.4$.

Watch for: The denominator is the condition group, not the entire population. Mutually exclusive nonzero-probability events are not independent.

Sampling, estimates and margin of error

$\text{estimated count} = \text{sample proportion} \times \text{population size}$

$\text{interval} = \text{estimate} \pm \text{margin of error}$

A representative random sample supports estimates about its population. A margin of error describes sampling uncertainty; larger comparable random samples tend to reduce it.

Example: An estimate of 52% with a 4-point margin gives 48% to 56%. A sample proportion 0.30 predicts about 600 of 2,000 people.

Watch for: A larger sample does not repair selection bias. A sample result is an estimate, not an exact population count; an interval is not a guarantee.

Study design and evaluating claims

Random sampling → population generalization

Random assignment → evidence for cause and effect in a well-designed experiment

Observational studies measure existing conditions; experiments assign treatments. Random assignment helps balance other factors between groups. Consider control groups, confounding, and the population actually sampled.

Example: Randomly assigning volunteers to two study methods can support a causal comparison for similar participants; recruiting only volunteers limits broad generalization.

Watch for: Random selection and random assignment solve different problems. A correlation from observation alone does not establish causation.

Geometry and Trigonometry

Lines, angles and polygons

straight angle = 180° ; full turn = 360° ; triangle angles sum to 180°

interior-angle sum of an n -sided polygon = $(n - 2)180^\circ$

Vertical angles are equal. With parallel lines, corresponding and alternate interior angles are equal; same-side interior angles sum to 180° . A triangle exterior angle equals the two remote interior angles combined.

Example: A triangle with angles 48° and 67° has third angle 65° . A regular hexagon has each interior angle $720^\circ/6 = 120^\circ$.

Watch for: Equal-looking angles are not enough: use stated parallel lines or marked relationships. Diagrams need not be to scale.

Triangle similarity and congruence

similar lengths scale by k ; areas by k^2 ; volumes of similar solids by k^3

Similar triangles have equal corresponding angles and proportional corresponding sides. AA proves similarity; SSS, SAS, ASA, and AAS can prove triangle congruence.

Example: Similar triangles have corresponding sides 4 and 10, so $k = 2.5$. A side of 6 corresponds to 15; an area of 8 corresponds to 50.

Watch for: Match corresponding vertices before writing a proportion. Equal areas do not imply congruent shapes.

Triangle facts and the Pythagorean theorem

$a^2 + b^2 = c^2$ for a right triangle; $|a - b| < c < a + b$

c is the hypotenuse opposite the right angle. The triangle inequality limits any third side. Equal sides in an isosceles triangle face equal angles.

Example: A right triangle with legs 6 and 8 has hypotenuse $\sqrt{100} = 10$. Sides 4 and 7 require the third side to be greater than 3 and less than 11.

Watch for: The Pythagorean theorem applies only to right triangles. The hypotenuse is the longest side.

Special right triangles

45°–45°–90°: $x, x, x\sqrt{2}$

30°–60°–90°: $x, x\sqrt{3}, 2x$

In the second ratio, x is opposite 30°, $x\sqrt{3}$ opposite 60°, and $2x$ is the hypotenuse. These relationships are available on the SAT reference sheet.

Example: If a 30°–60°–90° triangle has hypotenuse 12, its legs are 6 and $6\sqrt{3}$.

Watch for: Identify the angle opposite each side; do not assume x labels whichever side is given.

Right-triangle trigonometry

$\sin \theta = \text{opposite/hypotenuse}$; $\cos \theta = \text{adjacent/hypotenuse}$; $\tan \theta = \text{opposite/adjacent}$

For an acute angle in a right triangle, label sides relative to that angle. Also $\sin \theta = \cos(90^\circ - \theta)$ and $\sin^2 \theta + \cos^2 \theta = 1$.

Example: For a 3–4–5 triangle and the angle opposite side 3, $\sin \theta = 3/5$, $\cos \theta = 4/5$, and $\tan \theta = 3/4$.

Watch for: Opposite and adjacent switch when the reference angle changes. Use degrees or radians consistently with the prompt.

Area, perimeter and surface area

rectangle: $A = lw$; triangle: $A = bh/2$; parallelogram: $A = bh$

trapezoid: $A = (b_1 + b_2)h/2$

Perimeter adds boundary lengths. Height is perpendicular to the base. For composite figures, add component areas or subtract holes. Surface area adds the areas of all exposed faces.

Example: A trapezoid with bases 6 and 10 and height 4 has area 32. A $2 \times 3 \times 4$ box has surface area $2(6 + 8 + 12) = 52$.

Watch for: A slanted side is not necessarily a height. Area uses square units; perimeter uses length units.

Circle area, circumference and angles

$A = \pi r^2$; $C = 2\pi r = \pi d$; $d = 2r$

A central angle equals the measure of its intercepted arc in degrees. An inscribed angle is half the intercepted arc; an angle inscribed in a semicircle is a right angle. A tangent is perpendicular to the radius at the point of tangency.

Example: A circle of radius 5 has area 25π and circumference 10π . An inscribed angle intercepting an 80° arc is 40°.

Watch for: Using diameter as radius makes the area four times too large. An arc length is a distance, not an angle measure.

Arc length, sector area and radians

$$s = (\theta/360^\circ)2\pi r; \text{ sector area} = (\theta/360^\circ)\pi r^2$$

$$s = r\theta \text{ and sector area} = r^2\theta/2 \text{ when } \theta \text{ is in radians; } 180^\circ = \pi \text{ radians}$$

An arc and sector take the same fraction of the whole circle as their central angle. Convert degrees to radians by multiplying by $\pi/180$.

Example: A 60° sector of a radius-6 circle has arc length 2π and area 6π . The angle is $\pi/3$ radians.

Watch for: The formulas $s = r\theta$ and $A = r^2\theta/2$ require radians, not degrees.

Coordinate distance, midpoint and circles

$$\text{distance} = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$$

$$\text{midpoint} = ((x_1 + x_2)/2, (y_1 + y_2)/2)$$

$$(x - h)^2 + (y - k)^2 = r^2$$

A circle has center (h, k) and radius r . Complete the square in x and y to recover the center from expanded form. Distance comes from the Pythagorean theorem.

Example: $x^2 + y^2 - 6x + 4y = 12$ becomes $(x - 3)^2 + (y + 2)^2 = 25$: center $(3, -2)$, radius 5.

Watch for: Read radius as the square root of the right-hand constant. Signs inside the squares are opposite the center coordinates.

Volume of prisms, cylinders, pyramids, cones and spheres

$$\text{prism: } V = Bh; \text{ cylinder: } V = \pi r^2 h; \text{ pyramid: } V = Bh/3$$

$$\text{cone: } V = \pi r^2 h/3; \text{ sphere: } V = 4\pi r^3/3$$

B is base area and h is perpendicular height. A rectangular prism uses $B = lw$. Combine or subtract solid volumes when a shape has multiple components.

Example: A cylinder with radius 3 and height 10 has volume 90π . A cone with the same radius and height has volume 30π .

Watch for: Volume uses cubic units. Doubling every dimension multiplies volume by 8, not 2. Do not use slant height for a cone's volume.

Problem-solving toolkit

Translate, solve, interpret and check

Define → model → solve → check → answer the requested quantity

Write what each variable means. Convert units first, use the structure of the expression, and substitute your result back. Check positivity, integer constraints, denominator restrictions, and the context.

Example: If x is the number of packs but the question asks for total items and each pack has 12, the final answer is $12x$, not x .

Watch for: A mathematically valid root may be impossible as a negative length, a fractional number of people, or a time outside the model.

Use graphs and a calculator strategically

Roots solve $f(x) = 0$; intersections solve $f(x) = g(x)$

A graphing calculator can check roots, intersections, vertices, tables, and fitted models. Use algebra for exact values and parameter relationships. Estimate first and choose a window that shows the relevant inputs.

Example: To solve $x^2 = 2x + 3$, graph both sides and find $x = -1$ and 3 ; substitution confirms them exactly.

Watch for: A graph window can hide solutions. A rounded decimal near $1/3$ is not proof of an exact fraction. Confirm the calculator's angle mode for trigonometry.

Reference sheet versus understanding

Know what each symbol means and when the relationship applies

The standard SAT reference information includes common circle, rectangle, triangle and solid-volume formulas, special right triangles, the Pythagorean theorem, and angle totals. Algebra, percent reasoning, statistics, and interpreting models still require understanding.

Example: If given a diameter, first halve it before using $A = \pi r^2$. If a triangle's height is not stated, derive the perpendicular height before using $A = bh/2$.

Watch for: This guide covers the core content domains, not every possible wording or problem. Pair review with official practice and explanation of your own mistakes.

Scope and sources

The following official sources define the topic coverage. Explanations and examples in this guide are original; these links provide the official descriptions and access to further practice.

SAT Math overview

<https://satsuite.collegeboard.org/sat/whats-on-the-test/math/overview>

Algebra scope

<https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types/algebra>

Advanced Math scope

<https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types/advanced>

Problem-Solving and Data Analysis scope

<https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types/problem-solving>

Geometry and Trigonometry scope

<https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types/geometry-trigonometry>